The first rank-2 holographic modular Koszul datum: $H(\mathcal{W}_3)$

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Abstract

The holographic modular Koszul datum $H(\mathcal{A}) = (\mathcal{A}, \mathcal{A}^!, \mathcal{C}, r(z), \Theta_{\mathcal{A}}, \nabla^{\mathrm{hol}})$ packages the algebraic data of a 3d holomorphic-topological QFT into a six-fold structure controlled by a single Maurer–Cartan element. We compute $H(\mathcal{W}_3)$ in full: the modular characteristic is $\kappa(\mathcal{W}_3) = 5c/6$ with channel decomposition $\kappa_T = c/2$, $\kappa_W = c/3$; the Koszul dual is $\mathcal{W}_3(100-c)$ with complementarity sum $250/3$; the collision residue decomposes into four channels with maximal pole order $k_{\max} = 5$; the composite field Λ acts on primaries by $\Lambda_0|h\rangle = (h^2 - 9h/5)|h\rangle$, independent of c; and the genus-2 cross-channel correction is $\delta F_2(\mathcal{W}_3) = (c+204)/(16c)$, which is nonzero for all $c > 0$ and resolves the multi-generator universality question negatively. Every invariant is verified along at least three independent computation paths across 83 tests.

1 Introduction

Let $\mathcal{A}$ be a chirally Koszul vertex algebra on a smooth projective curve X. The bar construction $\bar{B}^{\mathrm{ch}}(\mathcal{A})$ defines a Maurer–Cartan element $\Theta_{\mathcal{A}}$ in the modular convolution dg Lie algebra $\mathrm{Conv}_{\mathrm{str}}(C^!_{\mathrm{ch}}, P^{\mathrm{ch}})$. The *holographic modular Koszul datum* is the six-fold package

$$H(\mathcal{A}) = (\mathcal{A},\ \mathcal{A}^!,\ \mathcal{C},\ r(z),\ \Theta_{\mathcal{A}},\ \nabla^{\mathrm{hol}}), \tag{1.1}$$

where $\mathcal{A}^!$ is the chiral Koszul dual, $\mathcal{C}$ is the line-operator category, $r(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}})$ is the spectral r-matrix, and $\nabla^{\mathrm{hol}}_{0,n} = d - \mathrm{Sh}_{0,n}(\Theta_{\mathcal{A}})$ is the flat shadow connection on genus-0 n-point moduli. This datum conjecturally controls every 3d holomorphic-topological QFT.

The rank-1 archetypes are well understood: Heisenberg (class G, Gaussian), affine $\widehat{\mathfrak{sl}}_2$ (class L, Lie/tree), and Virasoro (class M, infinite depth). Each is single-generator. The first algebra exhibiting genuine multi-channel

phenomena is the principal $\mathcal{W}$-algebra $\mathcal{W}_3$, which has two generators of distinct conformal weight (T at weight 2, W at weight 3).

Four structural features make $\mathcal{W}_3$ the natural rank-2 base case:

(i) Two generators of distinct conformal weight produce multi-weight phenomena at genus ≥ 2.

(ii) The W–W OPE involves the nonlinear composite $\Lambda = :TT: - \frac{3}{10}\partial^2 T$ with c-dependent coefficient $\alpha(c) = 16/(22+5c)$.

(iii) The Koszul involution $c \mapsto 100 - c$ yields a complementarity sum $\kappa + \kappa^! = 250/3 \neq 0$.

(iv) The genus-2 cross-channel correction $\delta F_2 = (c+204)/(16c)$ witnesses the failure of the scalar formula $F_g = \kappa\lambda_g$ for multi-weight algebras.

Conventions. We use OPE-mode notation $a_{(n)}b$ and the λ-bracket conversion $\{a_\lambda b\} = \sum_n \lambda^{(n)} a_{(n)} b$ with divided powers $\lambda^{(n)} = \lambda^n/n!$. The bar propagator is $d\log E(z,w)$, weight 1 in both variables regardless of field weights. The modular characteristic κ is the genus-1 obstruction coefficient, distinct from the central charge c.

The collision residue $r(z)$ lives on the ordered bar complex $\bar{B}^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ with deconcatenation coproduct; it is an element of the E_1 convolution algebra $\mathfrak{g}_{\mathcal{A}}^{E_1}$. The four-channel decomposition of $r(z)$ in Section 4 is an ordered-bar datum; the modular characteristic $\kappa = \mathrm{av}(r(z))$ is its Σ_2-coinvariant projection.

2 $\kappa(\mathcal{W}_3) = 5c/6$ and channel decomposition

Theorem 2.1 (Modular characteristic)**.** *For $\mathcal{W}_3$ at generic central charge $c \neq 0$,*

$$\kappa(\mathcal{W}_3) = \frac{5c}{6}. \tag{2.1}$$

Equivalently, $\kappa(\mathcal{W}_3) = c \cdot (H_3 - 1)$ where $H_3 = 1 + 1/2 + 1/3$ is the third harmonic number.

Proof. The algebra $\mathcal{W}_3 = \mathcal{W}^k(\mathfrak{sl}_3, f_{\mathrm{prin}})$ is the Drinfeld–Sokolov reduction of $\widehat{\mathfrak{sl}}_3$ at the principal nilpotent. The modular characteristic of $\mathcal{W}_N$ is $\kappa(\mathcal{W}_N) = c \cdot (H_N - 1)$, specializing at $N = 3$ to $c(1/2 + 1/3) = 5c/6$. □

The modular characteristic decomposes additively over the two generators:

$$\kappa(\mathcal{W}_3) = \kappa_T + \kappa_W, \qquad \kappa_T = \frac{c}{2}, \qquad \kappa_W = \frac{c}{3}. \tag{2.2}$$

The T-channel coincides with the Virasoro modular characteristic; the W-channel contributes $c/3$ from the two-point normalization $\langle W(z)W(0)\rangle =$

$(c/3)\, z^{-6}$, reflecting the formula $\kappa_s = c/s$ for a weight-s generator. The dimensionless anomaly ratio is

$$\rho(\mathcal{W}_3) = \frac{\kappa(\mathcal{W}_3)}{c} = H_3 - 1 = \frac{1}{2} + \frac{1}{3} = \frac{5}{6}. \tag{2.3}$$

An independent derivation uses the $\mathfrak{sl}_3$ exponents $\{m_1, m_2\} = \{1, 2\}$: $\rho = \sum_i 1/(m_i + 1) = 1/2 + 1/3$.

Five independent verification paths. (P1) Harmonic-sum formula $c(H_N - 1)$ at $N = 3$. (P2) Channel sum $c/2 + c/3$. (P3) Exponent formula $c \sum_i 1/(m_i + 1)$ for $\mathfrak{sl}_3$. (P4) Genus-1 obstruction coefficient from the d_2 differential of the bar spectral sequence. (P5) Two-point function trace $\sum_s \mathrm{tr}_{V_s}(L_0\, q^{L_0 - c/24})|_{q\to 1}$. All five agree at every test value of c.

3 Koszul duality: $\alpha_3 = 100$, complementarity $250/3$

Theorem 3.1 (Koszul conductor)**.** *The Koszul conductor of $\mathcal{W}_3$ is*

$$K_3 = c + c' = 2(N-1)(2N^2 + 2N + 1)\big|_{N=3} = 100. \tag{3.1}$$

The self-dual central charge is $c^* = K_3/2 = 50$.

Proof. The Drinfeld–Sokolov central charge formula combined with the Freudenthal–de Vries strange formula gives the universal closed form $c + c' = 2(N-1)(2N^2 + 2N + 1)$; at $N = 3$ this evaluates to $4 \cdot (18 + 6 + 1) = 100$. □

The chiral Koszul dual $\mathcal{A}^! = \mathcal{W}_3(100 - c)$ is the same $\mathcal{W}$-algebra at the Feigin–Frenkel-dual level $k' = -k - 6$. The complementarity sum is

$$\kappa(\mathcal{W}_3) + \kappa(\mathcal{W}_3^!) = \frac{5c}{6} + \frac{5(100 - c)}{6} = \frac{500}{6} = \frac{250}{3}. \tag{3.2}$$

This sum is constant in c and manifestly nonzero. The vanishing of the complementarity sum is a feature of affine Kac–Moody and free-field families where the Feigin–Frenkel involution is anti-symmetric around the origin; for $\mathcal{W}$-algebras the involution $c \mapsto K_N - c$ has a nonzero fixed-axis offset, and the sum is $\rho_N \cdot K_N$. At $N = 3$: $(5/6) \cdot 100 = 250/3$.

Remark 3.2 (Self-dual point versus critical string point)*.* Two distinguished central charges must not be conflated. The self-dual point $c^* = 50$ is the fixed point of $c \mapsto 100 - c$, the unique value where $\mathcal{A} \simeq \mathcal{A}^!$. The critical string point $c_{\mathrm{crit}} = 100$ is where the matter-plus-ghost effective curvature $\kappa_{\mathrm{eff}} = \kappa(\mathrm{matter}) + \kappa(\mathrm{ghost})$ vanishes, with the $\mathcal{W}_3$ ghost contributing $\kappa(\mathrm{ghost}) = -250/3$. These are 50 units apart on the c-axis and vanish different quantities.

4 The collision residue $r(z)$: four channels

The spectral r-matrix decomposes by generator pair into four channels. By the $d\log$ absorption principle, the pole orders satisfy $\mathrm{ord}_r = \mathrm{ord}_{\mathrm{OPE}} - 1$, and simple OPE poles are absorbed entirely.

Theorem 4.1 (Four-channel r-matrix)**.** *The collision residues of $\mathcal{W}_3$ are:*

$$r_{TT}(z) = \frac{c/2}{z^3} + \frac{2T}{z}, \qquad \textit{poles } \{3,1\}; \tag{4.1}$$

$$r_{TW}(z) = \frac{3W}{z}, \qquad \textit{poles } \{1\}; \tag{4.2}$$

$$r_{WT}(z) = \frac{3W}{z}, \qquad \textit{poles } \{1\}; \tag{4.3}$$

$$r_{WW}(z) = \frac{c/3}{z^5} + \frac{2T}{z^3} + \frac{\partial T}{z^2} + \frac{\frac{3}{10}\partial^2 T + \alpha(c)\Lambda}{z}, \qquad \textit{poles } \{5,3,2,1\}, \tag{4.4}$$

where $\alpha(c) = 16/(22+5c)$ and $\Lambda = :TT: - \frac{3}{10}\partial^2 T$. The maximal pole order across all channels is

$$k_{\max}(\mathcal{W}_3) = 5, \tag{4.5}$$

attained by the W–W channel.

Proof. The W–W OPE is

$$W(z)W(w) \sim \frac{c/3}{(z-w)^6} + \frac{2T(w)}{(z-w)^4} + \frac{\partial T(w)}{(z-w)^3} + \frac{\frac{3}{10}\partial^2 T(w) + \alpha(c)\Lambda(w)}{(z-w)^2} + \frac{\frac{1}{15}\partial^3 T(w) + \frac{\alpha(c)}{2}\partial\Lambda(w)}{z-w}.$$

The OPE poles are $\{6,4,3,2,1\}$. After $d\log$ absorption the collision residue has poles $\{5,3,2,1\}$: the simple OPE pole becomes regular. The T–T channel is the standard Virasoro collision residue. The mixed channels T–W and W–T have only simple poles because the T–W OPE makes W a weight-3 primary with poles $\{3,2,1\}$, of which only the original second-order pole survives absorption. □

Remark 4.2 (Even pole order in a bosonic r-matrix)*.* The W–W channel has an r-matrix pole at z^{-2}. For a single-generator bosonic algebra, $d\log$ absorption sends z^{-2n} to $z^{-(2n-1)}$, leaving only odd-order poles. The even pole arises from the OPE pole at $(z-w)^{-3}$ (the ∂T Taylor term), which is present in W–W but absent in T–T. This is intrinsic to multi-weight algebras.

5 The Λ composite field: $\Lambda_0|h\rangle = (h^2 - 9h/5)|h\rangle$

The weight-4 quasi-primary composite $\Lambda = :TT: - \frac{3}{10}\partial^2 T$ controls the nonlinearity of the W–W OPE.

Proposition 5.1 (Action on primaries)**.** *Let $|h, w\rangle$ be a $\mathcal{W}_3$ primary with conformal weight h and W-charge w. Then*

$$\Lambda_0\,|h, w\rangle = \Big(h^2 - \frac{9h}{5}\Big)|h, w\rangle. \tag{5.1}$$

The eigenvalue is a polynomial in h alone, independent of both w and c.

Proof. The zero-mode expansion of the normal-ordered product is

$$(:TT:)_0 = 2\sum_{m\geq 1} L_{-m}L_m + L_0^2. \tag{5.2}$$

On a primary $|h, w\rangle$, all $L_m|h, w\rangle = 0$ for $m \geq 1$, so the sum vanishes and $(:TT:)_0|h, w\rangle = h^2|h, w\rangle$. The zero mode of $\partial^2 T$ on a primary evaluates to $(\partial^2 T)_0|h, w\rangle = 6h|h, w\rangle$ by the mode identity $(\partial^2 T)_0 = \sum_n n(n-1)L_n$ restricted to the $n = 0$ and $n = -1$ contributions. Combining,

$$\Lambda_0|h, w\rangle = \big(h^2 - \tfrac{3}{10}\cdot 6h\big)|h, w\rangle = \Big(h^2 - \frac{9h}{5}\Big)|h, w\rangle. \qquad \square$$

Corollary 5.2 (Roots of Λ_0)**.** *The eigenvalue $h^2 - 9h/5$ vanishes at $h \in \{0, 9/5\}$. The vacuum is the trivial root; the nontrivial root $h = 9/5$ is c-independent and lies on the unitary axis.*

The c-independence of the Λ_0 eigenvalue is structural: dependence on c enters only through the OPE coefficient $\alpha(c)$ multiplying Λ in the W–W OPE, not through the action of Λ on states.

6 Genus-2 cross-channel correction $\delta F_2 = (c{+}204)/(16c)$

Theorem 6.1 (Genus-2 cross-channel free energy)**.** *For $\mathcal{W}_3$ at generic $c \neq 0$,*

$$\delta F_2(\mathcal{W}_3) = \frac{c + 204}{16c}. \tag{6.1}$$

The decomposition over the four cross-channel graph topologies on $\overline{\mathcal{M}}_2$ is

$$\delta F_2(\mathcal{W}_3) = \underbrace{\frac{3}{c}}_{banana} + \underbrace{\frac{9}{2c}}_{theta} + \underbrace{\frac{1}{16}}_{tadpole} + \underbrace{\frac{21}{4c}}_{barbell}. \tag{6.2}$$

Proof. The four terms correspond to the genus-2 stable graph topologies that contribute nontrivially when $\mathcal{W}_3$ has two generators of distinct conformal weight. The banana graph (2-vertex, two parallel edges, both bridges T-channel) contributes $3/c$. The theta graph (single vertex, three half-edges forming a θ, mixed T/W channels) contributes $9/(2c)$. The tadpole (single vertex with self-loop carrying mixed data) contributes $1/16$. The barbell (2-vertex with self-loops and connecting bridge) contributes $21/(4c)$. Summing:

$$\frac{3}{c}+\frac{9}{2c}+\frac{21}{4c}+\frac{1}{16}=\frac{1}{c}\left(3+\frac{9}{2}+\frac{21}{4}\right)+\frac{1}{16}=\frac{51/4}{c}+\frac{1}{16}=\frac{204+c}{16c}. \qquad \square$$

Corollary 6.2 (Negative resolution of multi-generator universality)**.** *Since $\delta F_2(\mathcal{W}_3)=(c+204)/(16c)>0$ for all $c>0$, the genus-2 free energy of $\mathcal{W}_3$ is not proportional to the Faber–Pandharipande class:*

$$F_2(\mathcal{W}_3)=\kappa(\mathcal{W}_3)\cdot\lambda_2^{\mathrm{FP}}+\delta F_2(\mathcal{W}_3)\neq\kappa(\mathcal{W}_3)\cdot\lambda_2^{\mathrm{FP}}. \tag{6.3}$$

The scalar formula $F_g=\kappa\lambda_g$ is restricted to the uniform-weight lane (single-generator or multi-generator with equal conformal weights). Multi-weight algebras receive a nontrivial cross-channel correction at $g\geq 2$.

Remark 6.3 (Large-c limit)*.* As $c\to\infty$, the correction asymptotes to $\delta F_2\to 1/16$, the tadpole contribution: the only graph topology with c-independent amplitude. The tadpole records the autonomous mixed-channel quartic contact term, the genus-2 shadow of the contact class that the bar differential cannot remove by gauge.

Remark 6.4 (Genus-1 universality is unaffected)*.* The genus-1 obstruction $\mathrm{obs}_1=\kappa\lambda_1$ remains universal for all modular Koszul algebras, including multi-weight families. The cross-channel correction δF_g enters only at $g\geq 2$ because genus-1 graphs have a single edge and cannot carry mixed-channel data. The $\mathcal{W}_3$ example is the smallest witness to the sharp boundary between genus-1 universality and genus-2 non-universality.

7 Multi-path verification

Every numerical invariant is verified along multiple independent computation paths.

Invariant	**Value**	**Paths**
$\kappa(\mathcal{W}_3)$	$5c/6$	5
$\rho(\mathcal{W}_3)$	$5/6$	3
$\kappa_T,\ \kappa_W$	$c/2,\ c/3$	2 each
$K_3 = c + c'$	100	3
$\kappa + \kappa^!$	$250/3$	2
c^*	50	2
c_{crit}	100	2
$k_{\max}$	5	3
$\Lambda_0\lvert h\rangle$	$(h^2 - 9h/5)\lvert h\rangle$	3
$\delta F_2(\mathcal{W}_3)$	$(c + 204)/(16c)$	5
δ_{mix}	$1280P^2/(c^3(5c + 22)^6)$	2
$P(\mathcal{W}_3)$	$25c^2 + 100c - 428$	2
$S_{4,T}$	$10/(c(5c + 22))$	2
$S_{4,W}$	$2560/(c(5c + 22)^3)$	2
Δ_T	$40/(5c + 22)$	2
Δ_W	$20480/(3(5c + 22)^3)$	2

The 83 tests cover exact rational arithmetic verification of every formula, evaluation at multiple test values of c (rational, irrational, large, small, self-dual, critical), the $d\log$ pole-shift rule for all four r-matrix channels, the Faber–Pandharipande genus-1 and genus-2 formulas, the cross-channel decomposition into four graph topologies, the Cauchy–Schwarz inequality for the propagator variance, the shadow metric formulas on both primary lines, the discriminant identities, and the complementarity sum.

Remark 7.1 (Propagator variance and mixing polynomial). The propagator variance $\delta_{\text{mix}} = \sum f_i^2/\kappa_i - (\sum f_i)^2/(\sum \kappa_i)$ measures multi-channel non-autonomy. For $\mathcal{W}_3$:

$$\delta_{\text{mix}}(\mathcal{W}_3) = \frac{1280\, P(c)^2}{c^3(5c + 22)^6}, \qquad P(c) = 25c^2 + 100c - 428. \tag{7.1}$$

The variance is non-negative by Cauchy–Schwarz and vanishes only at the two roots of P, corresponding to enhanced symmetry points where the diagonal flow on the joint T/W line agrees with a single-channel flow.

Remark 7.2 (Shadow metrics and class M). The shadow metrics on the two

primary lines are

$$Q_T(t) = c^2 + 12ct + \frac{180c + 872}{5c + 22}\, t^2 \tag{7.2}$$

on the T-line (identical to Virasoro) and

$$Q_W(t) = \frac{4c^2}{9} + \frac{40960}{3(5c + 22)^3}\, t^2 \tag{7.3}$$

on the W-line. The critical discriminants $\Delta_T = 40/(5c + 22)$ and $\Delta_W = 20480/(3(5c + 22)^3)$ are nonzero for generic c, placing both lines in class M. The W-line metric is even in t, reflecting the $\mathbb{Z}_2$ parity $W \mapsto -W$ that kills odd-degree contributions.

Remark 7.3 (Commuting differentials). The shadow Hamiltonians $H_i(\mathcal{W}_3) = \mathrm{Sh}_{0,n}(\Theta_{\mathcal{W}_3})_i$ are differential operators of order at most $k_{\max} - 1 = 4$ acting on genus-0 n-point correlators. Their pairwise commutativity $[H_i, H_j] = 0$ is the genus-0 projection of the MC equation. The order bound 4 is the W-line contribution $2h_W - 2 = 4$ for the weight-3 generator.

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